

RESEARCH ARTICLE

Coronal brain atlas in stereotaxic coordinates of the African spiny mouse, *Acomys cahirinus*

Marta Vitorino^{1,2,3}  | Sónia Simão^{1,2,4}  | João B. Moreira^{1,2}  |
 Joana Nogueira-Rodrigues^{5,6}  | Joana Silva^{1,2}  | Ana Sofia Lourenço^{1,2}  |
 Vítor Fernandes^{1,2,4}  | Monica M. Sousa⁵  | Gustavo Tiscornia^{1,3,7}  |
 Inês M. Araújo^{1,2,4,8} 

¹Centre for Biomedical Research (CBMR), University of Algarve, Faro, Portugal

²Faculty of Medicine and Biomedical Sciences, University of Algarve, Faro, Portugal

³Center for Marine Sciences (CCMAR), University of Algarve, Faro, Portugal

⁴Algarve Biomedical Center Research Institute (ABC-RI), University of Algarve, Faro, Portugal

⁵Nerve Regeneration Group, Instituto de Biologia Molecular e Celular (IBMC), Instituto de Investigação e Inovação em Saúde (i3S), University of Porto, Porto, Portugal

⁶Graduate Program in Molecular and Cell Biology, Instituto de Ciências Biomédicas Abel Salazar (ICBAS), University of Porto, Porto, Portugal

⁷Clínica Eugén, Barcelona, Spain

⁸Champalimaud Research Program, Lisbon, Portugal

Correspondence

Marta Vitorino, Centre for Marine Sciences, Campus de Gambelas, Edif.7, 8005-139 Faro, Portugal.

Email: mvtorino@ualg.pt

Inês M. Araújo, Algarve Biomedical Center Research Institute (ABC-RI), University of Algarve, Campus de Gambelas, Edif. 2, 8005-139 Faro, Portugal.

Email: imaraujo@ualg.pt

Funding information

national portuguese funding, Grant/Award Numbers: FCT, PPBI-POCI-01-0145-FEDER-022122; Fundação para a Ciência e Tecnologia, Grant/Award Number: PTDC/BIA-ANM/0697/2014; Santa Casa da Misericórdia, Grant/Award Number: MC-39-2019; Wings for Life - Spinal Cord Research Foundation, Grant/Award Number: WFL-PT-21/20; FCT PhD fellowships, Grant/Award Numbers: SFRH/BD/130770/2017, SFRH/BD/131565/2017

Abstract

The African spiny mouse (*Acomys cahirinus*) is an emerging model of mammalian epimorphic regeneration that has aroused the interest of the scientific community in the last decade. To date, studies on brain repair have been hindered by the lack of knowledge on the neuroanatomy of this species. Here, we present a coronal brain atlas in stereotaxic coordinates, which allows for three-dimensional identification and localization of the brain structures of this species. The brain of 12-week-old spiny mice was mapped in stereotaxic coordinates using cresyl violet-stained brain sections obtained from coronal cryosectioning of the brain after transcardial perfusion with fixative. The atlas is presented in 42 plates representing sections spaced 240 μ m apart. Stereotaxic coordinates were validated using both a model of Parkinsonian lesion of the striatum with 6-hydroxydopamine and labeling of the corticospinal tract in the spiny mouse spinal cord using AAV1/2-GFP intracortical injections. This work presents a new tool in *A. cahirinus* neurobiology and opens new avenues of research for the investigation of the regenerative ability of *A. cahirinus* in models of brain disorders.

KEYWORDS

Acomys cahirinus, atlas, brain, disease models, epimorphic regeneration, stereotaxic surgery

1 | INTRODUCTION

The African spiny mouse has a number of remarkable biological characteristics that have led to its emergence as a novel experimental model in several fields. The genus *Acomys* is composed of over 20 different species distributed in various regions of Africa and the Middle East. In the wild, it is mostly found in dry grasslands and arid rocky environments. Its common name refers to the hard spine-like hairs found on its back. Despite it being called a “mouse,” it belongs to the subfamily Deomyinae, which diverged from the genus *Mus* (subfamily Murinae) some 23 million years ago. *Acomys* is a social breeder with a relatively long gestation period (39 ± 1 days) that produces small litters (two to three) of very precocious pups. They are born fully developed, with full coats of fur, unfurled ears, and open eyes; indeed, they become mobile by 24 h and are quite active only a few days later (Pinheiro et al., 2018). This has led to *Acomys* being used as a model for organogenesis, fetal development, and perinatal biology (Dickinson et al., 2013; Ellery et al., 2013, 2016). For several decades in the 20th century, it was used as a model for diet-induced diabetes, as when fed high energy diets in laboratory colonies, the animals became obese and hyperglycemic (Shafir et al., 2006). Interest in *Acomys* for diabetes research waned around the time transgenesis was developed in *Mus* toward the beginning of the 80s. In 2012, *Acomys* was described to have a skin 20 times weaker than *Mus*. Presumably an antipredatory adaptation, relatively small shear forces result in massive skin wounds, with the animals losing large patches of both epidermis and the underlying dermis, leaving the underlying muscle fascia exposed (Seifert et al., 2012). Remarkably, the animals hardly bleed and seem resistant to infection. Even more remarkably, they have been shown to heal by epimorphic regeneration, resulting in reestablishment of the original tissue architecture. This phenotype extends to the ear pinna. When full-thickness wounds of 4 mm in diameter (comprising approximately 35% of the pinna surface) are inflicted, the animals form a proliferative blastema that starts growing inward until the original wound is completely closed. Histological examination reveals *de novo* formation of epidermis (including hair follicles and sebaceous glands), dermis, underlying dermis, adipose tissue, hyaline cartilage, a degree of muscle regeneration, and extensive angiogenesis and re-innervation (Gawriluk et al., 2016; Matias Santos et al., 2016). *Mus*, as most mammals, simply heals the original cut by fibrotic scarring. Shortly after wounding, *Acomys* forms a bona fide blastema and mounts a proliferative response that is sustained in time, leading to accumulation of a mass of tissue that then undergoes morphogenesis. In contrast, an initial reentry of local cells into the cell cycle is quickly abrogated in *Mus*. The different response of both species involves differential regulation of nuclear localization of tumor suppressors p21 and p27 (Gawriluk et al., 2016).

An early question was whether the remarkable regenerative phenotype observed in *Acomys* was limited to skin and ear pinna or was more widespread, involving other tissues and organs. A number of reports in the last decade point to the later: depending on the tissue or organ

and the type and degree of injury, *Acomys* shows resistance to injury or regeneration in several systems. Skin regeneration has been characterized by transcriptomics and proteomics, indicating a complex interplay between immune response, extracellular matrix, and species-specific cell responses (Brant et al., 2015, 2016; Maden, 2018; Yoon et al., 2020).

Macrophages have been shown to be crucial for the regenerative response (Simkin et al., 2017) and, in vitro, *Acomys* cells show resistance to reactive oxygen species (ROS)-induced cellular senescence (Saxena et al., 2019). Repeated injection of myotoxins into striated muscle results in muscle injury that is completely regenerated in 3 weeks postinjury in *Acomys*, but leads to increased adipose cell deposition in *Mus* muscle (Maden et al., 2018). A model of acute injury in the kidney shows a distinct resistance to ischemic injury correlating with a lack of fibrotic response (Okamura et al., 2018). Resistance to injury is also apparent after ischemic insult to the adult cardiac muscle, with reduced scar formation, enhanced angiogenesis, and cardiomyocyte expansion resulting in improved cardiac function compared to equally injured *Mus* (Koopmans et al., 2021). However, bona fide regeneration has not yet been shown in either kidney or heart injury (Koopmans et al., 2021; Okamura et al., 2018). In contrast, regeneration seems to occur in response to spinal cord injury. After hemi-crush of spinal cord, *Acomys* showed both molecular and immunohistochemical response evidencing reduced fibrosis and upregulation of growth factors and neurogenesis-related genes. Moreover, *Acomys* showed a more rapid bladder function compared with *Mus* after the injury, suggesting that *Acomys* can regenerate spinal cord lesions (Streeter et al., 2020). Nevertheless, extended studies need to be performed to show axonal regrowth across the damage site.

The injury resistance and regenerative capability in response to a range of insults in several tissues/organs points to a general mechanism governing fibrotic and regenerative response in *Acomys*. These observations beg the question of how *Acomys* would respond to central nervous system injury. Neurobiology of *Acomys* remains fundamentally unexplored, and particularly the lack of a stereotaxic brain atlas has limited studies on the ability of brain repair in this model. Recently, neurogenic niches were identified in *Acomys* brain with identical localization compared with *Mus* (Maden et al., 2021). Stereotaxic brain atlases of the mouse (*Mus musculus*) and rat (*Rattus norvegicus*) have been fundamental for the development of several rodent models of neurological disease and for a vast number of experimental approaches in neuroscience. Stereotaxic brain surgery is widely used in lesion or toxin models of disease, for electrode insertion for electrophysiological studies, microdialysis probes, genetic manipulation using viral vectors for gene silencing or disease modeling, or circuit tracers using fluorescent dyes or genetic reporters such as fluorescent proteins like GFP, as examples. Besides the commonly used laboratory rodents like the mouse and rat, other species have been used in neuroscience research. We have now developed a coronal brain atlas in stereotaxic coordinates of the 12-week-old *Acomys cahirinus*, which allows for future exploration of brain repair in this exciting new mammalian model of regeneration.

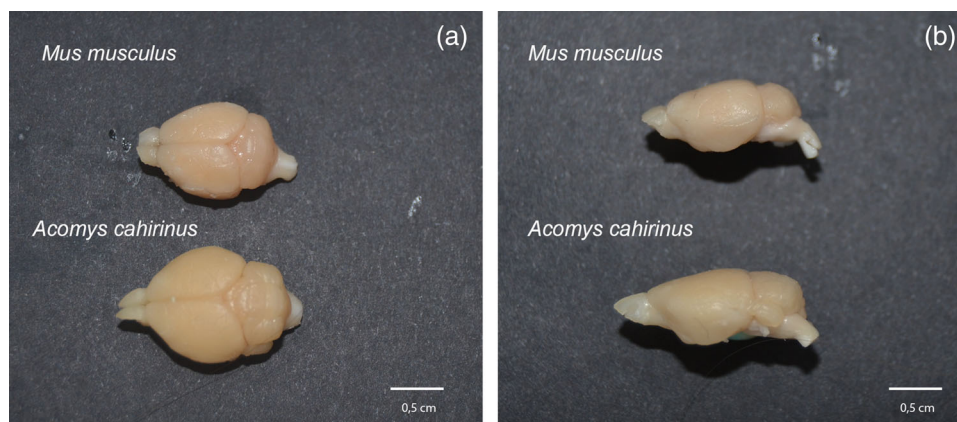


FIGURE 1 Comparison between *Acomys cahirinus* and *Mus musculus* brains. (a) Dorsal and (b) lateral views with the anterior part on the left. Scale bars: 0.5 cm

2 | MATERIALS AND METHODS

2.1 | Animals

Specimens of *A. cahirinus* were kept at the animal facility of the Centre for Biomedical Research at the University of Algarve, in a room with controlled temperature, on a 13/11 h light/dark cycle and with food (mix of seeds, with weekly supplementation of fresh fruits and vegetables) twice a week and water ad libitum, as described for this species (Pinheiro et al., 2018). All experiments were performed in accordance with European guidelines (2010/63/EU) for the care and use of laboratory animals, as well as Portuguese law (DL 113/2013). The procedures performed in mice have been reviewed and approved by the Animal Welfare Body of the University of Algarve and i3S and have been approved by the Direcção Geral de Alimentação e Veterinária (reference 0421/000/000/2016).

2.2 | Stereotaxic surgery and perfusion

The 12-week-old male spiny mice ($n = 5$, weight 35.4 ± 2.3 g) were anesthetized with 3% isoflurane in O_2 , and were placed in a stereotaxic apparatus (Stoelting, USA) in which the incisor bar was adjusted until the flat-skull position was achieved. After being placed in the correct position, bregma was located and was used as the origin of the system of coordinates. Two references were chosen, at the following coordinates (mm), reference site 1: anteroposterior (AP) 2.31, mediolateral (ML) -2.18, and dorsoventral (DV) -2.25; and reference site 2: AP -1.84, ML -2.75, and DV -2.28. The skull was perforated using a surgical drill and the brain was injected at the chosen coordinates using a Hamilton syringe with a 22-gauge needle.

Immediately after the surgical procedure, the animal was perfused transcardially with a saline solution (0.9% NaCl) followed by 4% paraformaldehyde in phosphate-buffered saline (PBS). The brain was carefully removed, postfixed in 4% paraformaldehyde overnight, and kept in 20% sucrose at 4°C until cryosectioning.

2.3 | Cryosectioning

The brain was embedded in Shandon Cryomatrix™ (Thermo Scientific, USA), frozen using Cytocool II (Thermo Scientific, USA), and 40- μ m coronal sections were cut using a cryostat (NX50 Cryostat, ThermoFisher Scientific, USA) at -20°C. Sections were divided in six different series, with a 240- μ m spacing between sections. The sections were directly mounted on Super Frost Plus microscope slides (VWR International, Germany) and air-dried overnight.

2.4 | Cresyl violet staining

Sections were stained on-slide with cresyl violet in every sixth section of 40 μ m. Briefly, the sections were washed for 2 min in distilled water, stained with cresyl violet (0.5% cresyl violet with 0.06% acetic acid) for 4 min, and then washed again for two more minutes in distilled water. Then, sections went through an ethanol gradient (70%, 95%, and absolute ethanol, for 5, 10, and 15 min, respectively) and were dipped two times in 100% xylene for 5 min each. Finally, the sections were cover-slipped with DPX Mountant for Histology (Sigma-Aldrich, USA) and maintained at room temperature until image acquisition.

2.5 | Image acquisition

The stained brain sections were visualized using an Axio Imager Z2 upright microscope (Zeiss, Germany). The images were obtained at a 12 $\times$ magnification, in a full section, using a Carl Zeiss AxioCam MRC. The AxioVision software (Zeiss, Germany) was used to photograph and convert the images. Subsequently, the obtained images were processed using Adobe Photoshop CC 2017 (Adobe Systems, USA) and illustrations were made with Canvas X (ACD Systems International, USA).

2.6 | Measurement of brain weight

To assess brain weight, fresh brains from 12-week-old male *A. cahirinus* ($n = 5$) were collected and weighed on a precision scale (Denver Instrument Company, USA).

2.7 | AP coordinates estimation

After the identification of the needle tract sections from reference site 1 and 2, it was possible to estimate the AP coordinates of the rest of the sections. First, the distance between the two reference points was calculated and divided by the number of sections between the two reference points. With these calculations, the rest of the coordinates were estimated.

2.8 | DV and ML coordinates estimation

The DV and ML coordinates were also estimated using the position of the needle tracts, from which a grid was created. First, we estimated the scale (in pixels) for ML coordinates by measuring the distance between the center of the lesion and with reference to the sagittal suture. Using the section where the first reference needle tract is visible, we measured the number of pixels between the needle tract and the sagittal suture. By converting this distance to micrometers and comparing it to the stereotaxic coordinate of our reference point, we could also calculate the shrinkage of the brain section, compared to the intact brain before surgery. Taking that into consideration, we were able to estimate the proper scale and create an ML coordinate axis. Based on the same principle, we were able to estimate the DV coordinates axis and scale in micrometers. To properly create the axis, we measured the distance from the bottom of the lesion upward, to the location of the skull surface, from which the axis starts, at 0 mm. These measurements were also performed for the second reference point to confirm the accuracy of the grid. This grid was applied to all sections of the brain map.

2.9 | 6-OHDA injection

The 12-week-old male spiny mice ($n = 5$) were anesthetized with 3% isoflurane in O_2 and were each placed in a stereotaxic apparatus (model 51900, Stoelting, USA), in which the incisor bar was adjusted until the flat-skull position was achieved. After being placed in the correct position, bregma was located and was used as the origin of the system of coordinates. Two references were chosen, at the following coordinates (mm), reference site 1: AP 2.31, ML 2.5, and DV -4.5; and reference site 2: AP 1.3, ML 3.4, and DV -4.5. The skull was perforated using a surgical drill and the brain was injected at the chosen coordinates using a Hamilton syringe with a 22-gauge needle with 8 μ g of 6-hydroxydopamine (6-OHDA, Catalog # H4381, Sigma-Aldrich, USA) in

0.9% NaCl and 0.2% ascorbate in a total of 2 μ l, at a rate of 0.5 μ l/min. Sham animals ($n = 5$) were subjected exactly to the same process, but injected with 2 μ l of 0.9% NaCl and 0.2% ascorbate at a rate of 0.5 μ l/min. After the surgical procedure, the animals were allowed to recover. After 8 weeks, the animals were perfused transcardially with a saline solution (0.9% NaCl) followed by 4% paraformaldehyde in PBS. The brain was carefully removed, postfixed in 4% paraformaldehyde overnight, and kept in 20% sucrose at 4°C until cryosectioning and stained for tyrosine hydroxylase (TH).

2.10 | Immunohistochemistry

To perform immunostaining against TH, the brains of *A. cahirinus* injected with 6-OHDA or saline (see procedure above) were cryosectioned in 30- μ m sections. The brain slices were divided in 10 series apart from each other 300 μ m. The brain sections were kept in Walter's Antifreeze solution (0.5 M Phosphate Buffer, Ethylene glycol, Glycerol) at 4°C until further use. Subsequently, a first treatment with hydrogen peroxide was performed to block peroxidase activity. Sections were then blocked with 5% goat serum in 0.25% Triton-PBS. TH detection was performed using the anti-TH antibody (1:5000, Sigma-Aldrich, USA) at 4°C for 48 h. A secondary antibody was used (biotinylated anti-rabbit IgG; 1:200, Vector Laboratories, USA). To amplify the antibody signal, avidin was conjugated with HRP for 3,3'-diaminobenzidine (DAB) conversion (1:50, Vector Laboratories, USA). Coverslips were mounted with DPX Mountant for Histology (Sigma-Aldrich, USA) in slides covered with 2% gelatin and maintained at room temperature until image acquisition in an Axio Imager Z2 upright microscope (Zeiss, Germany).

2.11 | CST labeling

For tracing of the motor descending corticospinal tract (CST), *A. cahirinus* were intracortically injected with an AAV1/2 vector leading to the expression of eGFP under the CMV promoter (AAV1.CMV.PI.eGFP.WPRE.bGH, 5.738×10^{13} GC/ml, Penn University). Animals were deeply anesthetized with continuous inhalation of isoflurane (Isovet, B Braun, 469860) and positioned in a stereotaxic device with motorized stereotaxic microinjector (Stoelting, USA). Following a midline incision, the bregma was exposed and four holes were drilled through the skull in the right sensorimotor cortex. The following coordinates were used, reference site 1: AP 1.35, ML -1.0, and DV 1.4; reference site 2: AP 1.35, ML -2.6, and DV 1.4; reference site 3: AP 0.63, ML -0.7, and DV 1.2; reference site 4: AP 0.63, ML -2.2, and DV 1.2. The injections were performed using a 10- μ l Hamilton syringe (Series 1700 Hamilton). Once the needle was at the appropriate depth, 0.3 μ l of virus were injected in each coordinate at 0.1 μ l/min. After the injection, the syringe was left for 2 min before being withdrawn to avoid leakage. Four weeks later, the animals were euthanized, tissue collected, and sectioned. Images were acquired using the IN Cell



FIGURE 2 *Acomys cahirinus* skull. (a) Lateral and (b) dorsal views of the skull with the anterior part on the left. (c) Dorsal view of the skull showing the bregma (B) and lambda (L), as well as the corresponding lines of connection. Scale bars: 0.5 cm

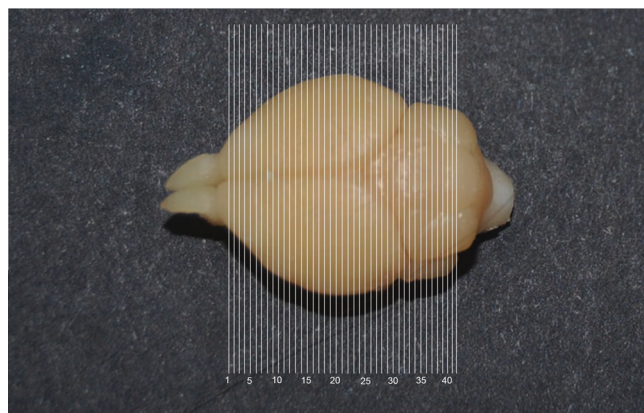


FIGURE 3 Anteroposterior location of the atlas plates on the *Acomys cahirinus* brain. Dorsal view. Distance between plates is 240 μ m

Analyzer 2000 microscope (GE Healthcare) at 20 $\times$ magnification with acquisition software v4.5 and analyzed in Fiji.

3 | RESULTS

In the present study, 12-week-old male *A. cahirinus* mice were used, which represent juvenile animals. Figure 1 shows a dorsal and lateral view of a fresh brain of *A. cahirinus* at 12 weeks. The brain of the spiny mouse is elongated, as has been described in the Mongolian gerbil (*Meriones unguiculatus*) (Radtke-Schuller et al., 2016). In contrast, the *Mus* brain has a more rounded shape (Figure 1) similar to that seen in the rat (*Rattus norvegicus*) (Paxinos & Watson, 2017), two of the most common rodents used as experimental models. At this stage, the brain of *A. cahirinus* weighs 0.762 ± 0.043 g ($n = 5$) on average.

Figure 2 shows the bregma and lambda of the *A. cahirinus* skull, as well as the corresponding lines of connection. These marks were used as the initial orientation to correctly position the mice in the stereotaxic apparatus. Two references were made in known coordinates by stereotaxic surgery and the sections obtained were ordered along AP axis.

Figure 3 shows the AP disposition of the atlas sections of the *A. cahirinus* brain. The distance between section was 240 μ m. *Acomys cahirinus* atlas, presented sequentially in Figure S1, contains 42 cresyl violet-stained coronal sections. Each of the 42 plates was assembled as a

combined image, using the Cresyl violet staining (left side of the plate) and the corresponding schematic draw delineating the identified structures on that plate (right side of the plate). Each plate contains the AP coordinates (in mm) from bregma and the DV and ML coordinates (in mm). A total of 600 different structures, encompassing all major brain areas characteristic of rodents, were identified across the 42 sections (Figure S1; Table 1). For comparison and labeling the anatomical structures of the *A. cahirinus* brain, the same nomenclature used for the Mongolian gerbil was adopted (Radtke-Schuller et al. 2016).

During the stereotaxic surgery, a needle was inserted into the specimen brain in two distinct sites to have two references (reference coordinate 1 and 2) relative to which the atlas' coordinates were estimated. The insertion of the needle produced a lesion, identifiable as a needle tract, in the brain parenchyma, as seen in sections 9 and 27. As can be observed in Figure 4a, the needle tract of reference coordinate 1 is visible on the left upper quadrant of the image, due to the needle being inserted at coordinates AP 2.31, ML -2.18, and DV -2.25. The brain regions affected by the needle tract of reference 1 were mainly the primary motor cortex, the cingulum, and the caudate putamen or striatum, which can be identified in more detail in Figure 4a1'. Figure 4b shows the needle tract of reference 2 with the coordinates AP -1.84, ML -2.75, and DV -2.28 (left upper quadrant of the image) with the main brain regions affected by this reference being the primary visual cortex, the deep cerebral white matter, the alveus, and the dorsal subiculum, which can be identified in more detail in Figure 4b1.

By evaluating the coronal sections of the spiny mouse brain atlas, it can be noticed that, in general, the gross morphology of the brain is similar to the morphology of other rodents, such as the mouse, rat, and gerbil (Dong, 2008; Paxinos & Watson, 2017; Radtke-Schuller et al., 2016). In particular, the dentate gyrus in the hippocampus of *A. cahirinus* (Figure 4, plates 18 and 19) presents a pronounced undulated granular cell layer, which is typical of the gerbil (Radtke-Schuller et al. 2016). Mice and rats do not have this peculiar format of the dentate gyrus, presenting instead a linear granular cell layer (Dong, 2008; Paxinos & Watson, 2017).

Next, we used the spiny mouse brain atlas described here to generate a set of coordinates with the aim of establishing a striatal lesion of dopaminergic neurons. Several authors described that the injection of 6-OHDA in the striatum of a mouse brain degenerates the dopaminergic neurons in both striatum and substantia nigra

TABLE 1 List of abbreviations

Abbreviation	Full name of the structure	Plate number
10Cb	10th cerebellar lobule (nodule)	42
1Cb	1st cerebellar lobule (lingula)	38, 39
2Cb	2nd cerebellar lobule	35–39
3Cb	3rd cerebellar lob	36–41
3n	Oculomotor nerve	28–30
3N	Oculomotor nucleus	30, 31
3PC	Oculomotor nucleus, parvicellular part	28–29
3V	3rd ventricle	11–26
4Cb	4th cerebellar lobule	36–39, 41
4n	Trochlear nerve	34–37
4N	Trochlear nucleus	32, 33
4Sh	Trochlear nucleus shell region	32, 33
4V	4th ventricle	36–42
5ADi	Motor trigeminal nucleus, anterior digastric part	38
5Cb	5th cerebellar lobule	36–42
5N	Motor trigeminal nucleus	37
5Sol	Trigeminal–solitary transition zone	40–42
5Tr	Trigeminal transition zone	37, 38
5TT	Motor trigeminal nucleus, tensor tympani part	36, 37
6Cb	6th cerebellar lobule	38–42
6n	Abducens nerve	38, 39
6N	Abducens nucleus	39
7DI	Facial nucleus, dorsal intermediate subnucleus	40–42
7DL	Facial nucleus, dorsolateral subnucleus	40–42
7DM	Facial nucleus, dorsomedial subnucleus	40–42
7L	Facial nucleus, lateral subnucleus	41, 42
7VI	Facial nucleus, ventral intermediate subnucleus	40–42
7VM	Facial nucleus, ventromedial subnucleus	40–42
7n	Facial nerve	38, 39
8n	Vestibulocochlear nerve	38–40
9Cb	9th cerebellar lobule	42
A1	Primary auditory field	17–25
A11	A11 dopamine cells	18–21
A13	A13 dopamine cells	16, 17
A5	A5 noradrenaline cells	37–39
AA	Anterior amygdaloid area	12, 13
AAF	Anterior auditory field	15–17
aca	Anterior commissure, anterior part	1–12
AcbC	Accumbens nucleus, core	5–9
AcbSh	Accumbens nucleus, shell	5–9
ACo	Anterior cortical amygdaloid nucleus	12–16
acp	Anterior commissure, posterior part	11–13
AD	Anterodorsal thalamic nucleus	14
AHA	Anterior hypothalamic area, anterior part	13
AHC	Anterior hypothalamic area, central part	14–16

(Continues)

TABLE 1 (Continued)

Abbreviation	Full name of the structure	Plate number
AHi	Amygdalohippocampal area	17–25
AHP	Anterior hypothalamic area, posterior part	15, 16
AI	Agranular insular cortex	1–14
alv	Alveus of the hippocampus	14–31
AM	Anteromedial thalamic nucleus	13–16
AMV	Anteromedial thalamic nucleus, ventral part	14
AngT	Angular thalamic nucleus	16, 17
ANS	Accessory neurosecretory nuclei	14
AOD	Anterior olfactory nucleus, dorsal part	1–3
AOL	Anterior olfactory nucleus, lateral part	1–3
AOM	Anterior olfactory nucleus, medial part	1–3
AOP	Anterior olfactory nucleus, posterior part	4–5
AOV	Anterior olfactory nucleus, ventral part	1, 2
AOVP	Anterior olfactory nucleus, ventroposterior part	1–4
APir	Amygdalopiriform transition area	20–27
APT	Anterior pretectal nucleus	27
APTD	Anterior pretectal nucleus, dorsal part	20–26
APTV	Anterior pretectal nucleus, ventral part	22–26
Aq	Aqueduct	27–35
Arc	Arcuate hypothalamic nucleus	16–25
asc7	Ascending fibers of the facial nerve	40
ASt	Amygdalostratial transition area	14–19
ATg	Anterior tegmental nucleus	33–34
Au1	Primary auditory cortex	15–25
AuD	Secondary auditory cortex, dorsal area	15–29
AuV	Secondary auditory cortex, ventral area	15–29
AV	Anteroventral thalamic nucleus	13
AVDM	Anterovent thalamic nucleus, dorsomedial part	16
B	Basal nucleus (Meynert)	12–17
BAC	Bed nucleus of the anterior commissure	12
Bar	Barrington's nucleus	36, 37
bic	Brachium of the inferior colliculus	28–33
BIC	Nucleus of the brachium of the inferior colliculus	28–30
BLA	Basolateral amygdaloid nucleus, anterior part	13–17
BLP	Basolateral amygdaloid nucleus, posterior part	14–23
BLV	Basolateral amygdaloid nucleus, ventral part	13, 14
BMA	Basomedial amygdaloid nucleus, anterior part	13, 14
BMP	Basomedial amygdaloid nucleus, posterior part	15–21
bsc	Brachium of the superior colliculus	24–27
CA1	Field CA1 of the hippocampus	15–29
CA2	Field CA2 of the hippocampus	15–23
CA3	Field CA3 of the hippocampus	14–26
cbw	Cerebellar white matter	35–42
cc	Corpus callosum	9–19
CeC	Central amygdaloid nucleus, capsular part	14–17

(Continues)

TABLE 1 (Continued)

Abbreviation	Full name of the structure	Plate number
CeL	Central amygdaloid nucleus, lateral division	14–16
CeM	Central amygdaloid nucleus, medial division	13–17
CEnt	Caudomedial entorhinal cortex	27–34
cg	Cingulum	6–26
CG	Central gray	36
CG1	Cingulate cortex, area 1	1–14
CG2	Cingulate cortex, area 2	8–14
CGA	Central gray, alpha part	37, 38
CGB	Central gray, beta part	37
CGG	Central gray, gamma part	38
CGO	Central gray, nucleus O	37
chp	Choroid plexus	13–25
cic	Commissure of the inferior colliculus	35
CIC	Central nucleus of the inferior colliculus	32–36
CI	Clastrum	3–4, 14
CL	Centrolateral thalamic nucleus	16–21
CLi	Caudal linear nucleus of the raphe	30–32
cll	Commissure of the lateral lemniscus	34, 35
CM	Central medial thalamic nucleus	14–21
CnF	Cuneiform nucleus	34–36
Cop	Copula of the pyramid	41, 42
cp	Cerebral peduncle	15–32
CPu	Caudate putamen (striatum)	6–19
CPO	Caudal periolivary nucleus	40
Crus1	Crus 1 of the ansiform lobule	38–42
Crus2	Crus 2 of the ansiform lobule	41, 42
csc	Commissure of the superior colliculus	26–29
CxA	Cortex–amygdala transition zone	12, 13
DA	Dorsal hypothalamic area	17–19
das	Dorsal acoustic stria	41, 42
DCDp	Dorsal cochlear nucleus, deep core	41
DCFu	Dorsal cochlear nucleus, fusiform layer	40, 41
DCI	Dorsal part of claustrum	6–13
DCIC	Dorsal cortex of the inferior colliculus	33–36
DCMo	Dorsal cochlear nucleus, molecular layer	40, 41
DMSP5	Dorsomedial spinal trigeminal nucleus	40, 41
dcw	Deep cerebral White matter	17–32
DEn	Dorsal endopiriform nucleus	3–23
DG	Dentate gyrus	18, 19
dhc	Dorsal hippocampal commissure	15–31
DI	Dysgranular insular cortex	4–14
Dk	Nucleus of Darkschewitsch	24–27
DLG	Dorsal lateral geniculate nucleus	17–25
dlo	Dorsal lateral olfactory tract	2, 3
DLO	Dorsolateral orbital cortex	1–3
DLPAG	Dorsolateral periaqueductal gray	28–35

(Continues)

TABLE 1 (Continued)

Abbreviation	Full name of the structure	Plate number
DM	Dorsomedial hypothalamic nucleus	17–19, 22–23
DMC	Dorsomedial hypothalamic nucleus, compact part	20, 21
DMD	Dorsomedial hypothalamic nucleus, dorsal part	20, 21
DMIC	Dorsomedial nucleus of the anterior colliculus	34–36
DMPAG	Dorsomedial periaqueductal gray	27–35
DMSp5	Dorsomedial spinal trigeminal nucleus	42
DMTg	Dorsomedial tegmental area	36, 37
DMV	Dorsomedial hypothalamic nucleus, ventral part	20, 21
DNLL	Dorsal nucleus of the lateral lemniscus	34, 35
DP	Dorsal peduncular cortex	3–7
DpG	Deep gray layer of the superior colliculus	24–34
DPGi	Dorsal paragigantocellular nucleus	40–42
DpWh	Deep white layer of the superior colliculus	26–34
DR	Dorsal raphe nucleus	30, 31
DRC	Dorsal raphe nucleus, caudal part	36, 37
DRD	Dorsal raphe nucleus, dorsal part	32–35
DRL	Dorsal raphe nucleus, lateral part	32–35
DRV	Dorsal raphe nucleus, ventral part	32–35
DS	Dorsal subiculum	24–29
DTg	Dorsal tegmental nucleus	37
DTgC	Dorsal tegmental nucleus, central part	36
DTgP	Dorsal tegmental nucleus, pericentral part	36
dtgx	Dorsal tegmental decussation	28–30
Dtr	Dorsal transition zone	4
DTT	Dorsal tenia tecta	1–7
dVNLL	Ventral nucleus of the lateral lemniscus, dorsal part	34, 35
E	Ependyma and subependymal layer	1–11
EAC	Sublenticular extended amygdala, central part	13
ec	External capsule	8–19
ECIC	External cortex of the inferior colliculus	31–38
Ect	Ectorhinal cortex	15–35
eml	External medullary lamina	15–19
Ent	Entorhinal cortex	35
Eth	Ethmoid thalamic nucleus	22, 23
Eve	Nucleus of origin of efferents of the vestibular nerve	39
EW	Edinger–Westphal nucleus	28–31
f	Fornix	12–26
F	Nucleus of the fields of Forel	22–25
FC	Fasciola cinereum	18, 19
fi	Fimbria of the hippocampus	10–19
Fl	Flocculus	35–40
fmi	Forceps minor of the corpus callosum	4–7
fmj	Forceps major of the corpus callosum	26–32
fr	Fasciculus retroflexus	16–23
Fr3	Frontal cortex, area 3	1–6

(Continues)

TABLE 1 (Continued)

Abbreviation	Full name of the structure	Plate number
g7	Genu of the facial nerve	38–40
Gca	Globular cell area, ventral cochlear nuclei	38–40
gcc	Genu of the corpus callosum	8
Gi	Gigantocellular reticular nucleus	39–42
GiA	Gigantocellular reticular nucleus, alpha part	40, 41
GI	Granular insular cortex	4–14
GP	Globus pallidus	12–17
GrC	Granule cell layer of cochlear nuclei	36–41
GrDG	Granular layer of the dentate gyrus	15–29
hbc	Habenular commissure	20–22
HDB	Nucleus of the horizontal limb of the diagonal band	9–13
hif	Hippocampal fissure	18–26
I	Intercalated nuclei of the amygdala	11–17
IAD	Interanterodorsal thalamic nucleus	14
IAM	Interanteromedial thalamic nucleus	14
ic	Internal capsule	11–19
icf	Intercrural fissure	41, 42
ICj	Islands of Calleja	6–10
icp	Inferior cerebellar peduncle (restiform body)	38–41
IEn	Intermediate endopiriform nucleus	3–12
IF	Interfascicular nucleus	27–31
IG	Indusium griseum	8–25
IL	Infralimbic cortex	4–7
IMD	Intermediodorsal thalamic nucleus	15–19
imvc	Intermedioventral thalamic commissure	17–19
IMG	Amygdaloid intramedullary gray	14–16
IntA	Interposed cerebellar nucleus, anterior part	39–41
InC	Interstitial nucleus of Cajal	24–31
InG	Intermediate gray layer of the superior colliculus	24–34
INLL	Intermediate nucleus of the lateral lemniscus	33–35
IntDL	Interposed cerebellar nucleus, dorsolateral hump	39–42
IntDM	Interposed cerebellar nucleus, dorsomedial crest	40–42
IntP	Interposed cerebellar nucleus, posterior part	41, 42
IntPPC	Interposed cerebellar nucleus, posterior parvicellular part	41, 42
InWh	Intermediate white layer of the superior colliculus	24–34
IPAC	Interstitial nucleus of the posterior limb of the anterior commissure	10–13
IPC	Interpeduncular nucleus, caudal subnucleus	30–33
IPD	Interpeduncular nucleus, dorsal subnucleus	31
IPF	Interpeduncular fossa	28, 29
IPI	Interpeduncular nucleus, intermediate subnucleus	31, 32
IPL	Interpeduncular nucleus, lateral subnucleus	30–32
IPR	Interpeduncular nucleus, rostral subnucleus	30, 31
IRt	Intermediate reticular nucleus	38–42
isRt	Isthmic reticular formation	32–35

(Continues)

TABLE 1 (Continued)

Abbreviation	Full name of the structure	Plate number
IVF	Interventricular foramen	13
KF	Kölliker–Fuse nucleus	36, 37
La	Lateral amygdaloid nucleus	14
LA	Lateroanterior hypothalamic nucleus	13
LaD	Lateral amygdaloid nucleus, dorsal part	16–19
LaV	Lateral amygdaloid nucleus, ventral part	16–19
LAcSh	Lateral accumbens shell	8, 9
Lat	Lateral (dentate) cerebellar nucleus	38–41
LatPC	Lateral cerebellar nucleus, parvicellular part	38–40
LC	Locus coeruleus	37
LDDM	Laterodorsal thalamic nucleus, dorsomedial part	16
LDTg	Laterodorsal tegmental nucleus	35–37
LDTgV	Laterodorsal tegmental nucleus, ventral part	36
LDVL	Laterodorsal thalamic nucleus, ventrolateral part	16, 17
LEnt	Lateral entorhinal cortex	17–34
lfp	Longitudinal fasciculus of the pons	33–36
Lhb	Lateral habenular nucleus	16, 20, 21
LhbL	Lateral habenular nucleus, lateral part	17–19
LhbM	Lateral habenular nucleus, medial part	17–19
II	Lateral lemniscus	33–36
LM	Lateral mammillary nucleus	24–27
LMol	Lacunosum moleculare layer of the hippocampus	17–29
LNTB	Lateral nucleus of the trapezoid body	37–39
LO	Lateral orbital cortex	1–5
lo	Lateral olfactory tract	1–13
LPAG	Lateral periaqueductal gray	27–35
LPBC	Lateral parabrachial nucleus, central part	36, 37
LPBCr	Lateral parabrachial nucleus, crescent part	37
LPBI	Lateral parabrachial nucleus, internal part	36, 37
LPGiA	Lateral paragigantocellular nucleus, alpha part	40–42
LPGiE	Lateral paragigantocellular nucleus, external part	40–42
LPLC	Lateral posterior thalamic nucleus, laterocaudal part	22–25
LPLR	Lateral posterior thalamic nucleus, laterorostral part	17–21
LPMC	Lateral posterior thalamic nucleus, mediocaudal part	22–26
LPMR	Lateral posterior thalamic nucleus, mediorostral part	17–23
LPO	Lateral preoptic area	10–13
LPtA	Lateral parietal association cortex	15–19
LSD	Lateral septal nucleus, dorsal part	8–13
LSI	Lateral septal nucleus, intermediate part	7–11
LSO	Lateral superior olive	38, 39
LSS	Lateral stripe of the striatum	11
LSV	Lateral septal nucleus, ventral part	8–11
LT	Lateral terminal nucleus of the accessory optic tract	24, 25
LTer	Lamina terminalis	12

(Continues)

TABLE 1 (Continued)

Abbreviation	Full name of the structure	Plate number
Lth	Lithoid nucleus	24, 25
LV	Lateral ventricle	6–21
LVe	Lateral vestibular nucleus	39–41
M1	Primary motor cortex	1–16
M2	Secondary motor cortex	1–16
MA3	Medial accessory oculomotor nucleus	26, 27
m5	Motor root of the trigeminal nerve	37
mch	Medial corticohypothalamic tract	13
mcp	Middle cerebellar peduncle	33–38
MCPC	Magnocellular nucleus of the posterior commissure	24–26
MCPO	Magnocellular preoptic nucleus	11–13
MD	Mediodorsal thalamic nucleus	14
MDC	Mediodorsal thalamic nucleus, central part	15–19
MDL	Mediodorsal thalamic nucleus, lateral part	15–19
MDM	Mediodorsal thalamic nucleus, medial part	15–19
ME	Median eminence	18–23
me5	Mesencephalic trigeminal tract	37
Me5	Mesencephalic trigeminal nucleus	31–37
MeAD	Medial amygdaloid nucleus, anterodorsal part	14
MeAV	Medial amygdaloid nucleus, anteroventral part	14
Med	Medial (fastigial) cerebellar nucleus	39–42
MedDL	Medial cerebellar nucleus, dorsolateral protuberance	41, 42
MedL	Medial cerebellar nucleus, lateral part	41, 42
MEnt	Medial entorhinal cortex	26–34
MePD	Medial amygdaloid nucleus, posterodorsal part	14–17
MePV	Medial amygdaloid nucleus, posteroventral part	15–17
MG	Medial geniculate nucleus	22, 23
MiTg	Microcellular tegmental nucleus	31, 32
mfb	Medial forebrain bundle	8–25
MGD	Medial geniculate nucleus, dorsal part	24–29
MGM	Medial geniculate nucleus, medial part	24–29
MGV	Medial geniculate nucleus, ventral part	24–29
Mhb	Medial habenular nucleus	14–21
ml	Medial lemniscus	15–42
ML	Medial mammillary nucleus, lateral part	26, 27
mlf	Medial longitudinal fasciculus	30–42
MM	Medial mammillary nucleus, medial part	26, 27
MnM	Medial mammillary nucleus, median part	26
MnPO	Median preoptic nucleus	10–12
MnR	Median raphe nucleus	32–36
MNTB	Medial nucleus of the trapezoid body	37–39
MoDG	Molecular layer of the dentate gyrus	15–30
MO	Medial orbital cortex	1–4
MoS	Molecular layer of the subiculum	30
mp	Mammillary peduncle	27–29

(Continues)

TABLE 1 (Continued)

Abbreviation	Full name of the structure	Plate number
MPA	Medial preoptic area	12, 13
MPB	Medial parabrachial nucleus	36, 37
MPBE	Medial parabrachial nucleus, external part	37
MPL	Paralemniscal nucleus, medial part	34, 35
MPO	Medial preoptic nucleus	12–13
MPT	Medial pretectal nucleus	22, 23
MPtA	Medial parietal association cortex	15–19
MRe	Mammillary recess of the 3 rd ventricle	26
mRt	Mesencephalic reticular formation	28–31
MS	Medial septal nucleus	8–11
MSO	Medial superior olive	37–39
mt	Mammillothalamic tract	18–25
MT	Medial terminal nucleus of the accessory optic tract	27
mtg	Mammillotegmental tract	24, 25
MTu	Medial tuberal nucleus	19–21
MVeMC	Medial vestibular nucleus, magnocellular part	38–42
MVePC	Medial vestibular nucleus, parvocellular part	38–42
Mx	Matrix region of the medulla	42
MZMG	Marginal zone of the medial geniculate	24–27
ns	Nigrostriatal bundle	14–25
Nv	Navicular nucleus of the basal forebrain	6, 7
Oca	Octopus cell area, ventral cochlear nucleus	39, 40
ocb	Olivocochlear bundle	38
och	Optic chiasm	12, 13
Op	Optic nerve layer of the superior colliculus	24–33
opt	Optic tract	14–25
OPT	Olivary pretectal nucleus	24, 25
Or	Oriens layer of the hippocampus	14–29
OT	Nucleus of the optic tract	22–27
OV	Olfactory ventricle (olfactory part of lateral ventricle)	1–5
p1PAG	p1 periaqueductal gray	22–26
p1Rt	p1 reticular formation	24–27
P5	Peritrigeminal zone	36–38
P7	Perifacial zone	40–42
Pa4	Paratrochlear nucleus	33, 34
Pa6	Paraabducens nucleus	38, 39
PaAP	Paraventricular hypothalamic nucleus, anterior parvocellular part	13
PaMM	Paraventricular hypothalamic nucleus, medial magnocellular part	16
PaPo	Paraventricular hypothalamic nucleus, posterior part	16
PaR	Pararubral nucleus	27–31
PaS	Parasubiculum	27–35
PaXi	Paraxiphoid nucleus of thalamus	16, 17
PBG	Parabigeminal nucleus	31–33
PBP	Parabrachial pigmented nucleus of the ventral tegmental area	24–30
pc	Posterior commissure	22–26

(Continues)

TABLE 1 (Continued)

Abbreviation	Full name of the structure	Plate number
PC	Paracentral thalamic nucleus	16–19
PCGS	Paracochlear glial substance	38
pcn	Precentral fissure	36–38
PCom	Nucleus of the posterior commissure	24–26
PCRt	Parvicellular reticular nucleus	38–42
pcuf	Preculminate fissure	37
PDR	Posterodorsal raphe nucleus	32–35
Pe	Periventricular hypothalamic nucleus	15–19
PeF	Perifornical nucleus	17–19
PeFLH	Perifornical part of lateral hypothalamus	18–23
PF	Parafascicular thalamic nucleus	20–23
PFI	Paraflocculus	35–42
pfs	Parafloccular sulcus	37–42
PH	Posterior hypothalamic nucleus	20–23
PHA	Posterior hypothalamic area	24–25
PHD	Posterior hypothalamic area, dorsal part	18–21
PIF	Parainterfascicular nucleus of the ventral tegmental area	28–30
PIL	Posterior intralaminar thalamic nucleus	24–27
Pir	Piriform cortex	1–25
PL	Paralemniscal nucleus	33–35
PLCo	Posterolateral cortical amygdaloid nucleus	14–17
PLH	Peduncular part of lateral hypothalamus	14–25
PLi	Posterior limitans thalamic nucleus	24–27
plf	Posterolateral fissure	37–42
PLV	Perilemniscal nucleus, ventral part	34, 35
pm	Principal mammillary tract	24, 25
PM	Paramedian lobule	41, 42
PMCo	Posteromedial cortical amygdaloid nucleus	17–23
PMD	Premammillary nucleus, dorsal part	24, 25
PMnR	Paramedian raphe nucleus	32–35
pms	Paramedian sulcus	42
PMV	Premammillary nucleus, ventral part	22–25
Pn	Pontine nuclei	32–35
PN	Paranigral nucleus of the ventral tegmental area	28–30
PnC	Pontine reticular nucleus, caudal part	37–39
PnO	Pontine reticular nucleus, oral part	32–36
PnR	Pontine raphe nucleus	36
PnV	Pontine reticular nucleus, ventral part	38, 39
Po	Posterior thalamic nuclear group	15–25
PoDG	Polymorph layer of the dentate gyrus	15–27
PoMn	Posteromedian thalamic nucleus	20, 21
Post	Postsubiculum	27–33
PoT	Posterior thalamic nuclear group, triangular part	24–27
PP	Peripeduncular nucleus	24–27
PPT	Posterior pretectal nucleus	24–27

(Continues)

TABLE 1 (Continued)

Abbreviation	Full name of the structure	Plate number
Pr	Prepositus nucleus	39–42
PR	Prerubral field	20–26
Pr5DM	Principal sensory trigeminal nucleus, dorsomedial part	37–39
Pr5VL	Principal sensory trigeminal nucleus, ventrolateral part	36–39
PrCnF	Precuneiform area	31–34
PreW	Pre-Edinger–Westphal nucleus	26, 27
PrC	Precommissural nucleus	20–25
prf	Primary fissure	36–42
PRh	Perirhinal cortex	15–35
PrL	Prelimbic cortex	1–7
PrS	Presubiculum	27–31
PS	Parastrial nucleus	12
psf	Posterior superior fissure	36–40
PSTh	Parasubthalamic nucleus	22, 23
PT	Paratenial thalamic nucleus	13, 14
Pta	Perifollicular tegmental area	32–36
Pte	Paraterete nucleus	17–19
PTg	Pedunculopontine tegmental nucleus	31–35
PtP	Parietal cortex, posterior area	16–21
PV	Paraventricular thalamic nucleus	14–16
PVA	Paraventricular thalamic nucleus, anterior part	13
PVP	Paraventricular thalamic nucleus, posterior part	17–21
py	Pyramidal tract	36–42
Py	Pyramidal cell layer of the hippocampus	14–29
Rad	Radiatum layer of the hippocampus	15–29
RAPir	Rostral amygdalopiriform area	15–19
Rbd	Rhabdoid nucleus	32
RCh	Retrochiasmatic area	16
Re	Reuniens thalamic nucleus	14–18
REth	Retroethmoid nucleus	24–25
rf	Rhinal fissure	1–34
Rh	Rhomboid thalamic nucleus	14–18
ri	Rhinal incisure	1–3
RI	Rostral interstitial nucleus of medial longitudinal fasciculus	22–25
RIP	Raphe interpositus nucleus	38–40
RLi	Rostral linear nucleus of the raphe	27–29
RMC	Red nucleus, magnocellular part	27–31
RMg	Raphe magnus nucleus	37–42
ROb	Raphe obscurus nucleus	42
RPa	Raphe pallidus nucleus	36–42
RPC	Red nucleus, parvicellular part	27–29
RR	Retrorubral nucleus	32, 33
RRe	Retrouniens area	20, 21
RRF	Retrorubral field	30–32
rs	Rubrospinal tract	35–42

(Continues)

TABLE 1 (Continued)

Abbreviation	Full name of the structure	Plate number
RSD	Retrosplenial dysgranular cortex	14–35
RSG	Retrosplenial granular cortex	33
RSGa	Retrosplenial granular cortex, a region	27–32
RSGb	Retrosplenial granular cortex, b region	22–32
RSGc	Retrosplenial granular cortex, c region	14–29
Rt	Reticular thalamic nucleus	14–19
RtTg	Reticulotegmental nucleus of the pons	34–37
RtTgP	Reticulotegmental nucleus of the pons, pericentral part	34–35
S1	Primary somatosensory cortex	12–19
S1BF	Primary somatosensory cortex, barrel field	9–16
S1DZ	Primary somatosensory cortex, dysgranular zone	5–7
S1DZO	Primary somatosensory cortex, oral dysgranular zone	7
S1FL	Primary somatosensory cortex, forelimb region	5–12
S1HL	Primary somatosensory cortex, hindlimb region	9–13
S1Sh	Primary somatosensory cortex, shoulder region	13
S1J	Primary somatosensory cortex, jaw region	4–9
S1Tr	Primary somatosensory cortex, trunk region	14–16
S1ULp	Primary somatosensory cortex, upper lip region	7–16
S2	Secondary somatosensory cortex	7–16
s5	Sensory root of the trigeminal nerve	36–39
Sag	Sagulum nucleus	35, 36
Sc	Scaphoid thalamic nucleus	22, 23
Sca	Spherical cell area, ventral cochlear nucleus	36–38
scc	Splenium of the corpus callosum	20–25
SCh	Suprachiasmatic nucleus	13
SCO	Subcommissural organ	23–26
scp	Superior cerebellar peduncle (brachium conjunctivum)	34–41
SG	Supragenulate thalamic nucleus	24–29
Sfi	Septofimbrial nucleus	12–13
SFO	Subfornical organ	13
SGe	Supragenual nucleus	38
SHi	Septohippocampal nucleus	7–12
SHy	Septohypothalamic nucleus	10–13
SIB	Substantia innominata, basal part	10–12
Sim	Simple lobule	38–41
simf	Simplex fissure	39–41
SLu	Stratum lucidum of the hippocampus	15–25
sm	Stria medullaris of the thalamus	13–19
SMV	Superior medullary velum	38, 39
SNC	Substantia nigra, compact part	24–32
SNR	Substantia nigra, reticular part	24–32
SO	Supraoptic nucleus	12–14
sol	Solitary tract	42
Sol	Nucleus of the solitary tract	41, 42
sox	Supraoptic decussation	14–23

(Continues)

TABLE 1 (Continued)

Abbreviation	Full name of the structure	Plate number
sp5	Spinal trigeminal tract	40–42
SP5I	Spinal trigeminal nucleus, interpolar part	41, 42
Sp5O	Spinal trigeminal nucleus, oral part	40–42
SPa	Subparaventricular zone of the hypothalamus	15, 16
SPF	Subparafascicular thalamic nucleus	20, 21
SPFPC	Subparafascicular thalamic nucleus, parvicellular part	20–23
SPN	Superior periolivary nucleus	37–39
SPTg	Subpeduncular tegmental nucleus	33–35
SpVe	Spinal vestibular nucleus	40–42
st	Stria terminalis	11–21
ST	Bed nucleus of the stria terminalis	10
StA	Strial part of the preoptic area	11
STIA	Bed nucleus of the stria terminalis, intraamygdaloid division	16, 17
STh	Subthalamic nucleus	19–23
STLP	Bed nucleus of the stria terminalis, lateral division, posterior part	11, 12
STLI	Bed nucleus of the stria terminalis, lateral division, intermediate part	12
STLV	Bed nucleus of the stria terminalis, lateral division, ventral part	11, 12
STMA	Bed nucleus of the stria terminalis, medial division, anterior part	11, 12
STMP	Bed nucleus of the stria terminalis, medial division, posterior part	13
STMV	Bed nucleus of the stria terminalis, medial division, ventral part	11, 12
str	Superior thalamic radiation	20–23
STr	Subiculum, transition area	30, 31
Su3	Supraoculomotor periaqueductal gray	28–32
Su3C	Supraoculomotor cap	28–30
Su5	Supratrigeminal nucleus	36, 37
Sub	Submedial thalamic nucleus	18, 19
SubB	Subbrachial nucleus	28–31
SubCA	Subcoeruleus nucleus, alpha part	37
SubCD	Subcoeruleus nucleus, dorsal part	36, 37
SubCV	Subcoeruleus nucleus, ventral part	36, 37
SubD	Submedial thalamic nucleus, dorsal part	15–17
SubG	Subgeniculate nucleus	20–23
SubI	Subincertal nucleus	17
SubV	Submedial thalamic nucleus, ventral part	15–17
SuG	Superficial gray layer of the superior colliculus	24–34
SuML	Supramammillary nucleus, lateral part	24–26
SuMM	Supramammillary nucleus, medial part	24–26
sumx	Supramammillary decussation	26
SuVe	Superior vestibular nucleus	38, 39
Te	Terete hypothalamic nucleus	20, 21
TeA	Temporal association cortex	30–32
TG	Tectal gray	24–27
ts	Tectospinal tract	32–42
TS	Triangular septal nucleus	12, 13

(Continues)

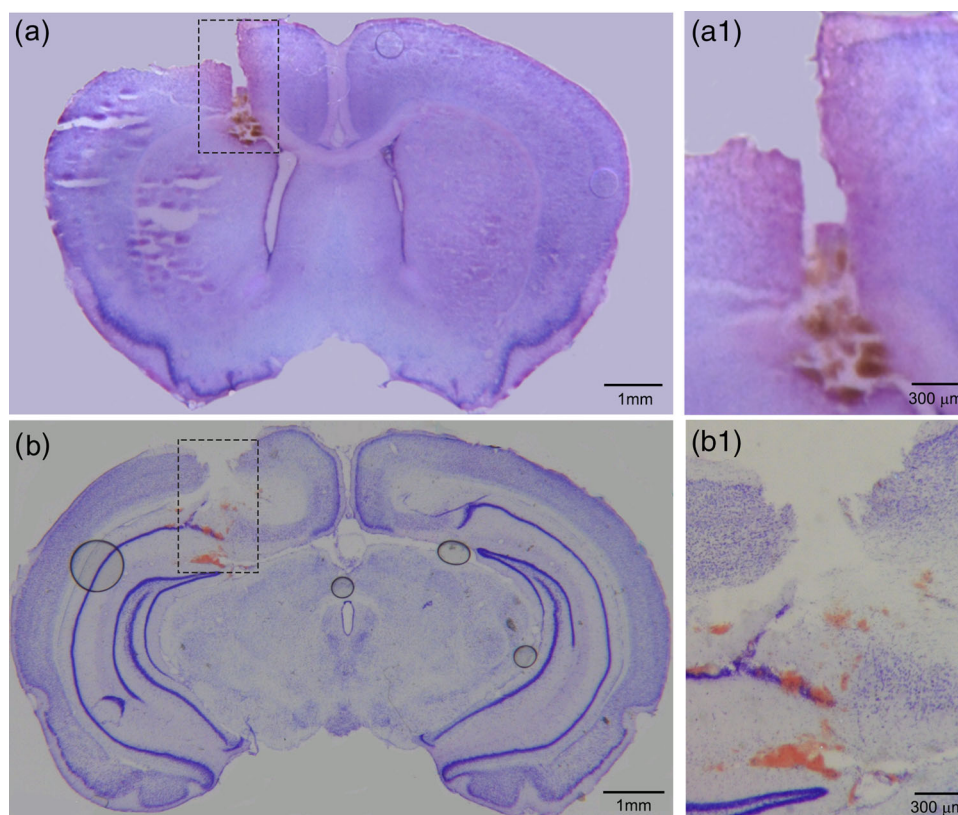
TABLE 1 (Continued)

Abbreviation	Full name of the structure	Plate number
tth	Trigeminothalamic tract	36, 37
Tu	Olfactory tubercle	5–11
TuLH	Tuberal region of lateral hypothalamus	15–21
tz	Trapezoid body	36–41
tzx	Decussation of the trapezoid body	37–39
V1	Primary visual cortex	20, 21, 33–35
V1B	Primary visual cortex, binocular area	22–32
V1M	Primary visual cortex, monocular area	22–32
V2L	Secondary visual cortex, lateral area	20–34
V2M	Secondary visual cortex, medial area	20–33
VA	Ventral anterior thalamic nucleus	14
VCA	Ventral cochlear nucleus, anterior part	36–39
VCI	Ventral part of claustrum	5–13
VCP	Ventral cochlear nucleus, posterior part	39, 40
VDB	Nucleus of the vertical limb of the diagonal band	8, 9
VeCb	Vestibulocerebellar nucleus	38–41
veme	Vestibulomesencephalic tract	38–40
VEen	Ventral endopiriform nucleus	12–17
vesp	Vestibulospinal tract	41
VG	Ventral geniculate nucleus	17–19, 24–26
VGMC	Ventral geniculate nucleus, magnocellular part	20–23
VGPC	Ventral geniculate nucleus, parvocellular part	20–23
vhc	Ventral hippocampal commissure	13, 14
VL	Ventrolateral thalamic nucleus	14–19
vlh	Ventrolateral hypothalamic tract	14
VLH	Ventrolateral hypothalamic nucleus	14
VLi	Ventral linear nucleus of the thalamus	24, 25
VLPA	Ventrolateral periaqueductal gray	30–35
VLPO	Ventrolateral preoptic nucleus	11, 12
VM	Ventromedial thalamic nucleus	14–19
VMH	Ventromedial hypothalamic nucleus	19–21
VMHC	Ventromedial hypothalamic nucleus, central part	18
VMHDM	Ventromedial hypothalamic nucleus, dorsomedial part	17, 18
VMHSh	Ventromedial nucleus of the hypothalamus shell	18–21
VMHVL	Ventromedial hypothalamic nucleus, ventrolateral part	18
VMPO	Ventromedial preoptic nucleus	11, 12
VNLL	Ventral nucleus of the lateral lemniscus	33
VNTB	Ventral nucleus of the trapezoid body	37–39
VO	Ventral orbital cortex	1–5
VP	Ventral pallidum	7–13
VPL	Ventral posterolateral thalamic nucleus	15–21
VPM	Ventral posteromedial thalamic nucleus	15–23
VPPC	Ventral posterior nucleus of the thalamus, parvocellular part	20, 21
VRe	Ventral reuniens thalamic nucleus	17–19
VS	Ventral subiculum	20–29

(Continues)

TABLE 1 (Continued)

Abbreviation	Full name of the structure	Plate number
vsc	Ventral spinocerebellar tract	36–42
VTA	Ventral tegmental area	30, 31
VTAR	Ventral tegmental area, rostral part	26, 27
VTg	Ventral tegmental nucleus	35
vtgx	Ventral tegmental decussation	27–30
VTM	Ventral tuberomammillary nucleus	24–26
VTT	Ventral tenia tecta	1–5
vVNLL	Ventral nucleus of the lateral lemniscus, ventral part	34–36
X	Nucleus X	40–42
Y	Nucleus Y	40
xscp	Decussation of the superior cerebellar peduncle	32–34
ZI	Zona incerta	15–17
ZIC	Zona incerta, caudal part	26, 27
ZID	Zona incerta, dorsal part	18–25
ZIR	Zona incerta, rostral part	14
ZIV	Zona incerta, ventral part	18–25
Zo	Zonal layer of the superior colliculus	24–34

**FIGURE 4** Cresyl violet staining of *Acomys cahirinus* brain sections, showing the needle tract. (a–a1') Reference 1 and (b–b1) reference 2. a1 and b1 (scale bar: 300 μ m) are amplification of a and b (scale bar: 1 mm), respectively

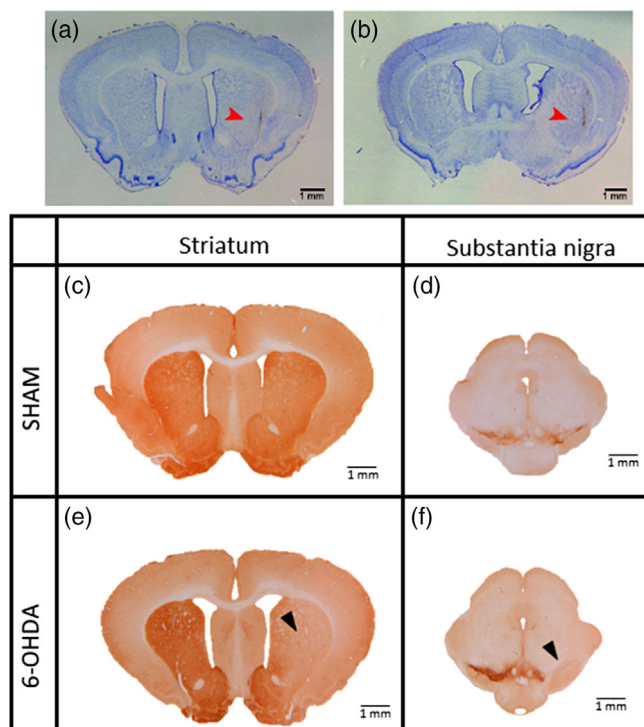


FIGURE 5 Loss of dopaminergic neurons in both striatum and substantia nigra induced by 6-OHDA injection. (a and b) Cresyl violet staining of *Acomys cahirinus* brain sections 4 weeks after 6-OHDA injection, showing the needle track (red arrowhead). (c–f) Immunohistochemistry against tyrosine hydroxylase in spiny mice brains injected with saline (c and d) or 6-OHDA (e and f) in the striatum. Black arrowheads indicate a decrease in dopaminergic neurons in both striatum (e) and substantia nigra (f) in 6-OHDA-injected animals. Scale bars: 1 mm

(Faull & Laverty, 1969; Przedborski et al., 1995). The injection of 6-OHDA in the mouse brain is a well-accepted Parkinson disease animal model (Betarbet et al., 2002; Schober, 2004). 6-OHDA was injected into two striatal sites at AP +2.31, ML +2.5, and DV −4.5 and AP +1.3, ML +3.4, and DV −4.5, from bregma of *A. cahirinus* atlas to produce a dopaminergic nigrostriatal lesion similar to what is performed in *Mus*. Figure 5 shows the needle tracts for the 2 injection sites in the striatum (Figures 5a and 5b, red arrowheads). Figure 5c–f represents a tyrosine hydroxylase staining for a saline-injected (sham c and d) or 6-OHDA-injected (e and f) *A. cahirinus* brain. In the group injected with 6-OHDA in the striatum, there was a notable decrease in the number of dopaminergic neurons (TH staining) on the injected side of the brain in both striatum (Figure 5e, black arrowhead) and substantia nigra (Figure 5f, black arrowhead). In the sham group, the TH staining was not affected, and no lesion was observed in the striatum or the substantia nigra (Figures 5c and 5d, respectively).

The *A. cahirinus* brain atlas was also used to define a set of coordinates intended to trace the motor descending corticospinal tract originating from pyramidal neurons at layer 5 of the brain. For that, AAV1/2-GFP were intracortically injected into the brain of *A. cahirinus* at the following coordinates, reference site 1: AP 1.35, ML −1.0, and DV 1.4;

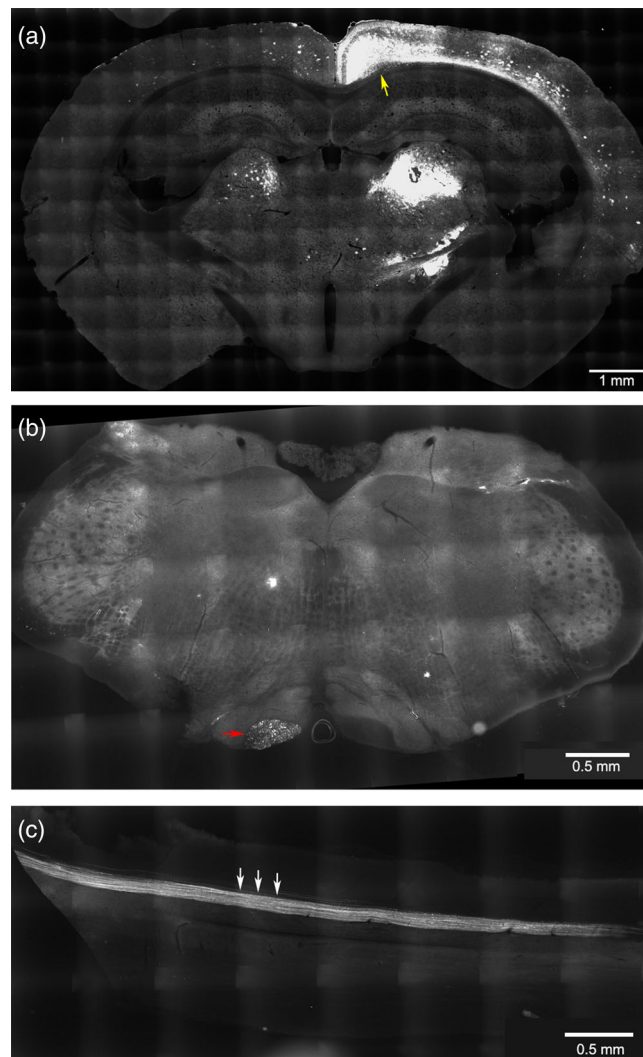


FIGURE 6 Descending corticospinal tract (CST) originated from pyramidal neurons. (a) GFP detected 4 weeks after injection of AAV1/2-GFP in the cortex (yellow arrow) and (b) brainstem (red arrow) of *Acomys cahirinus* brain. (c) GFP is also present in the white matter of a naïve spinal cord (white arrows). Scale bars: (a) 1 mm, (b and c) 0.5 mm

reference site 2: AP 1.35, ML −2.6, and DV 1.4; reference site 3: AP 0.63, ML −0.7, and DV 1.2; reference site 4: AP 0.63, ML −2.2, and DV 1.2. Following a 4-week postinjection period, we detected GFP-positive axons in the cortex of *A. cahirinus* (Figure 6a), suggestive of AAV anterograde axonal transport and transneuronal transduction. In line with these observations, GFP-positive fibers were also observed in the brainstem (Figure 6b). Furthermore, the correct anterograde transport of the AAV in the CST was confirmed by the presence of GFP-positive fibers in sagittal spinal cord sections of *A. cahirinus* throughout the white matter of the spinal cord (Figure 6c). The same coordinates used here to trace the motor descending corticospinal tract were also used by Nogueira-Rodrigues et al. (2022) to trace the CST of *A. cahirinus* subjected to a severe spinal cord injury.

4 | DISCUSSION

The value of a brain atlas is unquestionable in the field of neuroscience, constituting a fundamental tool required to perform precise stereotaxic surgery. The brain atlases of the main rodents used for experimental research are widely used (e.g., Dong, 2008; Hof et al., 2000; Paxinos & Watson, 2017; Radtke-Schuller et al., 2016; Swanson, 2018). However, the brain atlas of the *Acomys* species is not available and is a missing tool for this new regeneration model. The present study maps, for the first time, the brain of *A. cahirinus* in stereotaxic coordinates, making a coronal atlas of the brain of this species available to the scientific community. Many drugs and experimental approaches require local brain administration, which implies the knowledge of exact location of structures and brain nuclei in a precise system of coordinates. This is paramount for studies designed to establish lesion models, to administer toxins, dyes, or other substances intracranially, particularly in the case of drugs that do not cross the blood–brain barrier or even for studies aimed to manipulate gene expression using viral vectors.

Spiny mice are relatively long-lived rodents, with an average longevity near 5 years in controlled housing (Pinheiro et al., 2018; Weigl, 2005). Both males and female *Acomys* become sexually mature between 3 and 4 months (Pinheiro et al., 2018). Based on this, and while a 12-week-old mouse is a juvenile, it is already sexually mature. Using older mice also presents economic constraints of long-term housing costs, and 12 weeks is considered an adequate age for studies using juvenile–young adult individuals, especially for acute models of disease that make use of toxins or viral vectors for gene delivery. At 12 weeks of age, other common model species such as mice (*M. musculus*) and rat (*Rattus norvegicus*) are also frequently used in studies that require age-matched groups. For these reasons, 12 weeks of age was chosen to conduct the present study.

We found that the brain of *A. cahirinus* comprises the main regions and structures described for rodents such as the mouse, rat, or gerbil. The brain of a 12-week-old *A. cahirinus* weighs on average 0.762 ± 0.043 g ($n = 5$). By comparing this value with the brain weight of 16-week-old male C57BL/6J mice (0.420 ± 0.021 g [$n = 15$]) and with a 13-week-old male Sprague–Dawley rat (2.010 ± 0.110 g [$n = 862$]) (Piao, Liu et al. 2013), we estimate that the brain of *A. cahirinus* is 81% larger than the mouse and approximately 62% smaller than the rat. Wilkinson described that the brain of a 10-week-old Mongolian gerbil (*Meriones unguiculatus*) weighs on average 1.056 ± 8.000 g ($n = 5$) (Wilkinson 1986), which compares better with the data obtained by us for the spiny mouse brain, with only a 28% weight difference. Furthermore, our data demonstrate that in addition to weight, the brain of *A. cahirinus* resembles the brain of gerbil in features such as brain shape and the dentate gyrus format more closely than that of *Mus*. These similarities are consistent with the phylogenetic proximity between these two species. The spiny mice of the genus *Acomys* belong to the subfamily Deomyinae, which are a group of rodents from the large family Muroidea (Steppan & Schenk, 2017). From an evolutionary perspective, the Deomyinae subfamily is more closely related to the subfamily

Gerbillinae than to the subfamily Muridae (Chevret et al., 1993; Stepan & Schenk, 2017).

Tissue fixation can alter brain size after surgery, which might explain the difference between the theoretic distance between sections ($240 \mu\text{m}$) and the real distance ($\approx 230 \mu\text{m}$). In fact, we detected a shrinkage of 19% of the brain sections after performing histological procedures as described in this work, compared to the intact brain before surgery. This tissue shrinkage results from the histological procedure, mostly due to the paraformaldehyde fixation followed by dehydration in sucrose, and has been previously reported in other brain atlases (Radtke-Schuller et al., 2016). With this experimental consideration, in the present study we were able to estimate the AP, DV, and ML coordinates of every sixth section of a 12-week-old *A. cahirinus* brain.

In recent years, the spiny mouse has emerged as a novel experimental model for mammalian regeneration. This model has an extraordinary resistance to injury and the ability to regenerate a wide range of tissues and systems (Gawriluk et al., 2016; Maden, 2018; Maden et al., 2018; Matias Santos et al., 2016; Okamura et al., 2018; Seifert et al., 2012; Streeter et al., 2020). Surprisingly, little is known concerning the *Acomys* central nervous system. Very recently, Maden et al. (2021) examined stem cell populations in the *A. cahirinus* brain in comparison to *M. musculus* and described that these cell niches are localized in the same brain regions in both animals. In the present atlas, we observe a peculiar arrangement of the dentate gyrus in the brain of *A. cahirinus*, with an undulated granular cell layer. Interestingly, apart from the gerbil that also presents a similar undulated configuration of the dentate gyrus (Radtke-Schuller et al. 2016), this format was not reported so far in the brain of other rodents. In the study of Maden et al. (2021), the undulated dentate gyrus of *A. cahirinus* brain (contrasting to the linear format of the *Mus*) is also apparent, although this is not so evident as in our study, probably due to the fact that these authors performed sagittal sections in contrast to the coronal sections performed here.

We have validated the present atlas, by performing two specific sets of experiments: first, we produced a 6-OHDA lesion in the dopaminergic terminal of the striatum, a widely used model of Parkinson's disease in mice and rats. By defining a new pair of coordinates, we were able to establish a 6-OHDA-induced striatal lesion in the *A. cahirinus* brain. We found that this procedure induced a well-established striatal lesion in the *A. cahirinus* brain, with evident dopaminergic neurodegeneration and a clear loss of the TH staining in the 6-OHDA-injected mice, contrasting to the absence of changes in coloration in the sham group. Second, we successfully labeled the corticospinal tract with AAV-GFP intracortically injected into the brain of *A. cahirinus*, indicative of an effectively mapping of *A. cahirinus* stereotaxic coordinates. Moreover, Nogueira-Rodrigues et al. (2022) also validated this atlas by using this tool to estimate coordinates to label the corticospinal tract of spinal cord-injured animals. Altogether, these experiments successfully validated the stereotaxic coordinates used to assemble the atlas of the *A. cahirinus* brain and will allow further studies in this species.

Overall, we believe that the atlas presented here will be of great value to neuroscience research in the *Acomys* model and will permit

further innovative studies on the spiny mouse nervous system. Due to the impressive regenerative ability of this species, this atlas might open new research avenues, allowing to respond to several unanswered questions concerning different central nervous system pathologies.

ACKNOWLEDGMENTS

We acknowledge the Light Microscopy Unit and Animal House of CBMR-UAlg. The Microscopy Unit is partially supported by national portuguese funding FCT: PPBI-POCI-01-0145-FEDER-022122. Inês Araújo is funded by the Algarve Biomedical Center (ABC) and Loule Municipality. This work was supported by Fundação para a Ciência e Tecnologia (PTDC/BIA-ANM/0697/2014), Santa Casa da Misericórdia (MC-39-2019) and Wings for Life - Spinal Cord Research Foundation (WFL-PT-21/20, project number 234). Joana Silva and Joana Rodrigues are recipients of FCT PhD fellowships SFRH/BD/130770/2017 and SFRH/BD/131565/2017, respectively.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in the Supporting Information of this article

PEER REVIEW

The peer review history for this article is available at <https://publons.com/publon/10.1002/cne.25329>.

ORCID

Marta Vitorino  <https://orcid.org/0000-0001-8563-8975>

Sónia Simão  <https://orcid.org/0000-0002-0245-0633>

João B. Moreira  <https://orcid.org/0000-0002-7736-844X>

Joana Nogueira-Rodrigues  <https://orcid.org/0000-0001-8918-7753>

Joana Silva  <https://orcid.org/0000-0003-0306-2146>

Ana Sofia Lourenço  <https://orcid.org/0000-0003-2882-6026>

Vítor Fernandes  <https://orcid.org/0000-0001-5179-2122>

Monica M. Sousa  <https://orcid.org/0000-0002-4524-2260>

Gustavo Tiscornia  <https://orcid.org/0000-0002-1841-5330>

Inês M. Araújo  <https://orcid.org/0000-0002-2438-0111>

REFERENCES

- Betarbet, R., Sherer, T. B., & Greenamyre, J. T. (2002). Animal models of Parkinson's disease. *Bioessays*, 24(4), 308–318. <https://doi.org/10.1002/bies.10067>
- Brant, J. O., Lopez, M. C., Baker, H. V., Barbazuk, W. B., & Maden, M. (2015). A comparative analysis of gene expression profiles during skin regeneration in *Mus* and *Acomys*. *PLoS ONE*, 10(11), e0142931. <https://doi.org/10.1371/journal.pone.0142931>
- Brant, J. O., Yoon, J. H., Polvadore, T., Barbazuk, W. B., & Maden, M. (2016). Cellular events during scar-free skin regeneration in the spiny mouse, *Acomys*. *Wound Repair and Regeneration*, 24(1), 75–88. <https://doi.org/10.1111/wrr.12385>
- Chevret, P., Denys, C., Jaeger, J. J., Michaux, J., & Catzeflis, F. M. (1993). Molecular evidence that the spiny mouse (*Acomys*) is more closely related to gerbils (Gerbillinae) than to true mice (Murinae). *Proceedings of the National Academy of Sciences of the United States of America*, 90(8), 3433–3436. <https://doi.org/10.1073/pnas.90.8.3433>
- Dickinson, H., Ireland, Z. J., Larosa, D. A., O'Connell, B. A., Ellery, S., Snow, R., & Walker, D. W. (2013). Maternal dietary creatine supplementation does not alter the capacity for creatine synthesis in the newborn spiny mouse. *Reproductive Sciences*, 20(9), 1096–1102. <https://doi.org/10.1177/1933719113477478>
- Dong, H. W. (2008). The Allen reference atlas: A digital color brain atlas of the C57Bl/6J male mouse. John Wiley & Sons Inc.
- Ellery, S. J., Ireland, Z., Kett, M. M., Snow, R., Walker, D. W., & Dickinson, H. (2013). Creatine pretreatment prevents birth asphyxia-induced injury of the newborn spiny mouse kidney. *Pediatric Research*, 73(2), 201–208. <https://doi.org/10.1038/pr.2012.174>
- Ellery, S. J., LaRosa, D. A., Kett, M. M., Della Gatta, P. A., Snow, R. J., Walker, D. W., & Dickinson, H. (2016). Dietary creatine supplementation during pregnancy: A study on the effects of creatine supplementation on creatine homeostasis and renal excretory function in spiny mice. *Amino Acids*, 48(8), 1819–1830. <https://doi.org/10.1007/s00726-015-2150-7>
- Faull, R. L., & Lavery, R. (1969). Changes in dopamine levels in the corpus striatum following lesions in the substantia nigra. *Experimental Neurology*, 23(3), 332–340. [https://doi.org/10.1016/0014-4886\(69\)90081-8](https://doi.org/10.1016/0014-4886(69)90081-8)
- Gawriluk, T. R., Simkin, J., Thompson, K. L., Biswas, S. K., Clare-Salzler, Z., Kimani, J. M., Kiama, S. G., Smith, J. J., Ezenwa, V. O., & Seifert, A. W. (2016). Comparative analysis of ear-hole closure identifies epimorphic regeneration as a discrete trait in mammals. *Nature Communications*, 7, 11164. <https://doi.org/10.1038/ncomms11164>
- Hof, P. R., Young, W. G., Bloom, F. E., Belinchenko, P. V., & Celi, M.-R. (Eds.). (2000). *Comparative cytoarchitectonic atlas of the C57BL/6 and 129/Sv mouse brains with CDROM*. Elsevier.
- Koopmans, T., van Beijnum, H., Roovers, E. F., Malhotra, D., Tomasso, A., Boeter, J., Versteeg, D., van Rooij, E., & Bartscherer, K. (2021). Ischemic tolerance and cardiac repair in the African spiny mouse. *npj Regenerative Medicine*, 6, 78.
- Maden, M. (2018). Optimal skin regeneration after full thickness thermal burn injury in the spiny mouse, *Acomys cahirinus*. *Burns: Journal of the International Society for Burn Injuries*, 44(6), 1509–1520. <https://doi.org/10.1016/j.burns.2018.05.018>
- Maden, M., Brant, J. O., Rubiano, A., Sandoval, A. G. W., Simmons, C., Mitchell, R., Collin-Hooper, H., Jacobson, J., Omairi, S., & Patel, K. (2018). Perfect chronic skeletal muscle regeneration in adult spiny mice, *Acomys cahirinus*. *Scientific Reports*, 8(1), 8920. <https://doi.org/10.1038/s41598-018-27178-7>
- Maden, M., Serrano, N., Bermudez, M., & Sandoval, A. G. W. (2021). A profusion of neural stem cells in the brain of the spiny mouse, *Acomys cahirinus*. *Journal of Anatomy*, 238(5), 1191–1202. <https://doi.org/10.1111/joa.13373>
- Matias Santos, D., Rita, A. M., Casanellas, I., Brito Ova, A., Araujo, I. M., Power, D., & Tiscornia, G. (2016). Ear wound regeneration in the African spiny mouse *Acomys cahirinus*. *Regeneration*, 3(1), 52–61. <https://doi.org/10.1002/reg.2.50>
- Nogueira-Rodrigues, J., Leite, S. C., Pinto-Costa, R., Sousa, S. C., Luz, L. L., Sintra, M. A., Oliveira, R., Monteiro, A. C., Pinheiro, G. G., Vitorino, M., Silva, J. A., Simão, S., Fernandes, V. E., Provaznik, J., Benes, V., Cruz, C. D., Safronov, B. V., Magalhães, A., Reis, C. A., & Sousa, M. M. (2022). Rewired glycosylation activity promotes scarless regeneration and functional recovery in spiny mice after complete spinal cord transection. *Developmental Cell*, 57(4), 440.e7–450.e7. <https://doi.org/10.1016/j.devcel.2021.12.008>
- Okamura, D. M., Brewer, C. M., Wakenight, P., Bahrami, N., Bernardi, K., Tran, A., Olson, J., Shi, X., Piliponsky, A. M., Nelson, B. R., Beier, D. R., Millen, K. J., & Majesky, M. W. (2018). Scarless repair of acute and chronic kidney injury in African Spiny mice (*Acomys cahirinus*). *bioRxiv*, <https://doi.org/10.1101/315069>
- Paxinos, G., & Watson, C. (2017). *The rat brain in stereotaxic coordinates: Compact*. (7th ed.). Elsevier.

- Piao, Y., Liu, Y., & Xie, X. (2013). Change trends of organ weight background data in Sprague Dawley rats at different ages. *Journal of Toxicologic Pathology*, 26(1), 29–34. <https://doi.org/10.1293/tox.26.29>
- Pinheiro, G., Prata, D. F., Araujo, I. M., & Tiscornia, G. (2018). The African spiny mouse (*Acomys* spp.) as an emerging model for development and regeneration. *Laboratory Animals*, 52(6), 565–576. <https://doi.org/10.1177/0023677218769921>
- Przedborski, S., Levivier, M., Jiang, H., Ferreira, M., Jackson-Lewis, V., Donaldson, D., & Togasaki, D. M. (1995). Dose-dependent lesions of the dopaminergic nigrostriatal pathway induced by intrastratial injection of 6-hydroxydopamine. *Neuroscience*, 67(3), 631–647. [https://doi.org/10.1016/0306-4522\(95\)00066-r](https://doi.org/10.1016/0306-4522(95)00066-r)
- Radtke-Schuller, S., Schuller, G., Angenstein, F., Grosser, O. S., Goldschmidt, J., & Budinger, E. (2016). Brain atlas of the Mongolian gerbil (*Meriones unguiculatus*) in CT/MRI-aided stereotaxic coordinates. *Brain Structure and Function*, 221(Suppl. 1), 1–272. <https://doi.org/10.1007/s00429-016-1259-0>
- Saxena, S., Vekaria, H., Sullivan, P. G., & Seifert, A. W. (2019). Connective tissue fibroblasts from highly regenerative mammals are refractory to ROS-induced cellular senescence. *Nature Communications*, 10(1), 4400. <https://doi.org/10.1038/s41467-019-12398-w>
- Schober, A. (2004). Classic toxin-induced animal models of Parkinson's disease: 6-OHDA and MPTP. *Cell and Tissue Research*, 318(1), 215–224. <https://doi.org/10.1007/s00441-004-0938-y>
- Seifert, A. W., Kiama, S. G., Seifert, M. G., Goheen, J. R., Palmer, T. M., & Maden, M. (2012). Skin shedding and tissue regeneration in African spiny mice (*Acomys*). *Nature*, 489(7417), 561–565. <https://doi.org/10.1038/nature11499>
- Shafir, E., Ziv, E., & Kalman, R. (2006). Nutritionally induced diabetes in desert rodents as models of type 2 diabetes: *Acomys cahirinus* (spiny mice) and *Psammomys obesus* (desert gerbil). *ILAR Journal*, 47(3), 212–224. <https://doi.org/10.1093/ilar.47.3.212>
- Simkin, J., Gawriluk, T. R., Gensel, J. C., & Seifert, A. W. (2017). Macrophages are necessary for epimorphic regeneration in African spiny mice. *eLife*, 6, e24623. <https://doi.org/10.7554/eLife.24623>
- Steppan, S. J., & Schenk, J. J. (2017). Muroid rodent phylogenetics: 900-species tree reveals increasing diversification rates. *PLoS ONE*, 12(8), e0183070. <https://doi.org/10.1371/journal.pone.0183070>
- Streeter, K. A., Sunshine, M. D., Brant, J. O., Sandoval, A. G. W., Maden, M., & Fuller, D. D. (2020). Molecular and histologic outcomes following spinal cord injury in spiny mice, *Acomys cahirinus*. *Journal of Comparative Neurology*, 528(9), 1535–1547. <https://doi.org/10.1002/cne.24836>
- Swanson, L. W. (2018). Brain maps 4.0 - Structure of the rat brain: An open access atlas with global nervous system nomenclature ontology and flatmaps. *The Journal of Comparative Neurology*, 526(6), 935–943. <https://doi.org/10.1002/cne.24381>
- Weigl, R. (2005). *Longevity of mammals in captivity; From the Living Collections of the world*. E. Schweizerbart'sche.
- Wilkinson, F. (1986). Eye and brain growth in the Mongolian gerbil (*Meriones unguiculatus*). *Behavioural Brain Research*, 19(1), 59–69. [https://doi.org/10.1016/0166-4328\(86\)90048-3](https://doi.org/10.1016/0166-4328(86)90048-3)
- Yoon, J. H., Cho, K., Garrett, T. J., Finch, P., & Maden, M. (2020). Comparative proteomic analysis in scar-free skin regeneration in *Acomys cahirinus* and scarring *Mus musculus*. *Scientific Reports*, 10(1), 166. <https://doi.org/10.1038/s41598-019-56823-y>

SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

How to cite this article: Vitorino, M., Simão, S., Moreira, J. B., Nogueira-Rodrigues, J., Silva, J., Lourenço, A. S., Fernandes, V., Sousa, M. M., Tiscornia, G., & Araújo, I. M. (2022). Coronal brain atlas in stereotaxic coordinates of the African spiny mouse, *Acomys cahirinus*. *Journal of Comparative Neurology*, 530, 2215–2237. <https://doi.org/10.1002/cne.25329>